



Department of Computer Science and Engineering
BRAC University
CSE423 (Computer Graphics), Quiz: 2
Set: A, Full Marks: 20, Time: 30 minutes
Thursday 13th March, 2025

20

Student Number:

Name:

Section:

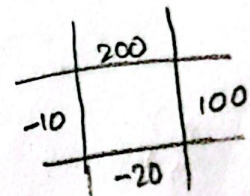
Signature of Student:

There are 4 questions in this script in 2 pages. Please make sure that you got all of them.
Answer in the space below each question. Extra papers shall be neither supplied nor are
you allowed to use any.

Consider the following for all outcodes in this quiz:

3	2	1	0
above	below	right	left

Region Code



1. Consider a rectangular viewing region (with sides parallel to the axes) bounded by the diagonal points (100, -20) and (-10, 200). (8)

Comment "Trivial Accept", "Trivial Reject" or "Neither" for each line, according to the Cohen-Sutherland Algorithm.

Line	Outcode - A	Outcode - B	Comment
A(20, 101) to B(-10, -20)	0000	0000	Trivial Accept ✓
A(-50, -30) to B(-11, 201)	0101	1001	Trivial Reject ✓
A(-15, 10) to B(150, -25)	0001	0110	Neither = Partial ✓ or reject
A(1000, 5000) to B(0, -5000)	1010	0100	Neither = Partial ✓ or reject

2. Why can not a point with the following outcode exist in the Cohen-Sutherland Algorithm: 1101 (2)

Because it logically doesn't make sense for a point to be both Above and Below the view region at the same time. ✓

It has to be either above or below and either left or right. or inside.

3. Fill up the following table (write only 1 point per cell):

(4)

	Cohen-Sutherland	Cyrus-Beck
Pros	Less calculations. = Faster + Fast trivial accept	Not limited to just rectangles.
Cons	Only limited to rectangles + variable iterations based on actual defn	Limited to polygons with obtuse angle.

4. Consider the clipping region to be bounded by the points (-10,-20) and (100,200). And the line segment to be drawn from (0,300) to (300,0). Fill up the empty spaces.

(6)

• $x_0 = 0$ $y_0 = 300$ $x_1 = 300$ $y_1 = 0$

• $D = (300, -300)$

$x_{\min} = -10$

$x_{\max} = 100$

$y_{\min} = -20$

$y_{\max} = 200$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_e	t_L
Left	(-1, 0)	-300	-0.033	PE	0	1
Right	(1, 0)	300	0.333	PL	0	0.333
Bottom	(0, -1)	300	1.067	PL	0	0.333
Top	(0, 1)	-300	0.333	PE	0.333	0.333

What is your final decision about the line segment according to the Cyrus-Beck Algorithm?

Here, $t_e = t_L \therefore$ The clipping segment is a single point.

$P(t_e) = P(t_L) = P(0.333) = (x_0 + t(x_1 - x_0), y_0 + t(y_1 - y_0))$

$\therefore P(0.333) = (100, 200)$

(Ans.)

$\frac{1}{3} = 0.333$